Why did Medicare ACOs reset their benchmarks early? A pre-registered analysis of early renewal in the Medicare Shared Savings Program

Samuel Thomas Harrold

October 2026

Independent researcher
ORCID: 0009-0008-4596-6921
Correspondence: Samuel Thomas Harrold, samuel.harrold@gmail.com

Preregistration: OSF Registries, doi:10.17605/OSF.IO/C53W9, registered October 26, 2026, before any outcome data were read.
Keywords: accountable care organizations; Medicare Shared Savings Program; benchmarks; alternative payment models; preregistration; difference-in-differences; Bayesian analysis
License: CC BY 4.0

Abstract

Importance: Medicare Shared Savings Program (MSSP) benchmarks are reset at each new agreement from an accountable care organization's (ACO's) recent spending, adjusted toward its region. ACOs could end an agreement early to reset sooner; to the author's knowledge, which ACOs did so has not been studied.

Objective: To test, under a preregistered plan, whether ACOs that renewed early stood to gain more than others from the regional part of a benchmark reset, and whether their spending growth differed afterward.

Design, Setting, and Participants: Preregistered retrospective cohort study of the 334 MSSP ACOs whose agreements began July 1, 2019 or January 1, 2020, using CMS public use files (2016-2024). The plan, decision rules and code were registered on OSF before any outcome data were read, with a code-level lock on the outcome files. The benchmark a reset would produce was rebuilt

from the regulation (42 CFR 425.601) and validated against benchmarks CMS published for 101 other ACOs before use.

Exposures: For the choice analyses, pre-decision characteristics: the regional pull (case-mix-adjusted regional spending relative to the ACO's current benchmark; primary), the own-spending component of a reset and the BASIC risk level. For the spending analysis, early renewal into a January 1, 2022 agreement.

Main Outcomes and Measures: Early renewal (29 ACOs, compared with 277 that stayed), analyzed by group comparisons with Bayesian bootstrap intervals and permutation tests and by a Bayesian logistic model with measurement error tested by direct simulation; and per-capita spending growth in 2022-2024, by difference-in-differences with a preregistered equivalence condition on pre-trends.

Results: The rebuilt benchmark tracked CMS's (Spearman $\rho = 0.94$; median absolute error 1.6%). Relative to their current benchmarks, early renewers' case-mix-adjusted regional spending was 3.4 percentage points higher than stayers' (95% credible interval [CrI], 1.7 to 5.4; $P < .001$). Early renewers were also beating their benchmarks by more (own-spending component, −2.1 points; 95% CrI, −3.8 to −0.5), and by the approximation the total expected gain from a reset did not differ (−0.5 points; 95% CrI, −2.0 to 0.8). Among 206 BASIC-track ACOs, early renewal was associated with a higher regional pull (log-odds difference per 2 SD, 1.34; 95% CrI, 0.44 to 2.30) and a higher, two-sided BASIC level (1.34; 95% CrI, 0.38 to 2.31). Spending growth did not differ detectably (−0.4%; 95% CrI, −4.0% to 2.5%; $P = .78$); the equivalence condition did not hold, so this is not read as an effect.

Conclusions and Relevance: ACOs that renewed early more often had regions that spent more relative to their benchmarks and were more often at higher BASIC levels, for which renewal opened a 75% sharing rate. By the study's approximation, a reset would not on average have raised their benchmarks. These are associations under the 2022 rules; benchmarks set from 2024 credit prior savings.

Introduction

In the Medicare Shared Savings Program (MSSP), an accountable care organization (ACO) shares in savings when the spending of its assigned beneficiaries falls below a benchmark.[1] At the start of each agreement period the benchmark is reset from the ACO's own spending in the 3 preceding years and adjusted part of the way toward the spending of its region.[2] Evaluations have found modest savings from the program, growing with tenure and larger in physician-group ACOs.[3–5] Whether selection, of clinicians, beneficiaries or whole ACOs, shapes those estimates has been examined with

CC BY 4.0

differing results: one analysis found no evidence of favorable risk selection in the original program,[6] while another found that accounting for the nonrandom exit of high-cost clinicians removed the estimated savings,[7] and risk-adjustment rules may lead ACOs to drop high-risk beneficiaries.[8]

Benchmarks are central to that selection. Rebasing to an ACO's own recent spending weakens its incentive to save, and under earlier weights it rewarded higher spending just before a new agreement.[9] ACOs' spending varies widely relative to their regions even after risk adjustment, so moving benchmarks toward regional averages shifts them unevenly.[10] After regional spending entered benchmarks in 2017, participation shifted toward providers whose spending was low relative to their region,[11] ACOs whose effective benchmarks fell became more likely to leave,[12] and bonuses rose for ACOs across the efficiency spectrum.[13] Survival in the program has been associated with earning bonuses, higher benchmarks and one-sided risk,[14] and larger ACOs, those with supporting organizations and better performers were more likely to switch to two-sided risk.[15] The 2019 Pathways to Success rules, which shortened the time ACOs could stay in one-sided risk, were followed by more exits than entries,[16] by ACO leaders reassessing potential savings against losses,[17] and by departures with projected costs to Medicare.[18] In a related model, most ACO Investment Model ACOs exited once downside risk was required.[19]

The same rules allowed an ACO to end its agreement early and begin a new one, which resets the benchmark early.[1] For an ACO in the BASIC track, renewal was also the only way into the ENHANCED track before its agreement ended, because the 2018 rule bars moving from BASIC to ENHANCED within an agreement period;[20] ENHANCED shares 75% of savings, against at most 50% in BASIC.[1] Of the 334 ACOs whose agreements began July 1, 2019 or January 1, 2020, 29 ended them early and began new agreements on January 1, 2022.[21] Which ACOs used the option bears on how benchmark rules sort ACOs, but to the author's knowledge it has not been studied.

Two motives predict early renewal. Under a **reset motive**, an ACO renews because its new benchmark would be higher. Under a **sharing-rate motive**, a BASIC-track ACO renews to reach ENHANCED. The savings rate cannot separate them: an ACO beating its benchmark would lose on the own-spending part of a reset, yet may gain on the regional part if its region spends more. This study therefore took as its primary test the **regional pull**, a measure of how far an ACO's case-mix-adjusted regional spending sat above its current benchmark, which contains no term for the ACO's own benchmark-year spending. It asked whether early renewers stood to gain more from the regional part of a reset than ACOs that stayed, which characteristics were associated with early renewal, and whether early renewers' spending growth differed afterward. The analysis plan, hypotheses, decision rules and code were registered before any outcome value was read.[22] Associations with pre-decision characteristics bear on the motives but cannot establish them, and no test here can separate the two motives.

CC BY 4.0

Methods

Study design and preregistration

This is a preregistered retrospective cohort study of ACO-level public data, reported following the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guideline[23] (checklist in Supplement 1). The analysis plan and the code commit that implements it were registered on OSF Registries on October 26, 2026, 13:22 UTC, with the plan's SHA-256 hash recorded.[22,24] Before registration only the enrollment files, the CMS data dictionaries, the regulation text and the column names (not the values) of the outcome files were used. The registry's own timestamp, rather than the repository's commit dates, establishes that order. The code also enforced it: every data file is hashed before it is read, and any file whose hash matches a performance-year results file, a county spending file or the county assignment file could not be opened until the registration record existed. The pipeline was written and tested on synthetic data with planted effects before registration. Analyses ran in stages that each refused to start until the outputs they depend on existed: the power analysis was committed before any confirmatory estimate, the robustness results (which depend only on validation and power) were committed before the confirmatory results and read only together with them, and every results file was committed before its estimates were read.

Data sources

Four CMS public use files were used, all aggregate at the ACO or county level:[21,25–27] MSSP ACO enrollment files for PY2016-PY2024, performance-year (PY) financial and quality results for PY2016-PY2024 (PY2023 and PY2024 only for the spending analysis), county-level expenditure and risk-score data on assignable beneficiaries for 2019-2021, and counts of ACO-assigned beneficiaries by county for 2021. The 40 files downloaded on October 22, 2026, each recorded with its size, SHA-256 hash and URL, also include files kept from the design stage that no analysis reads; an archived copy of the snapshot is on OSF.[28] Cells with counts of 1 to 10 are suppressed in these files.[29]

Population and groups

The base population was the 334 ACOs whose PY2021 enrollment record showed an agreement start of July 1, 2019 (189 ACOs) or January 1, 2020 (145 ACOs). Both cohorts' agreements end December 31, 2024, so an ACO from either cohort with a January 1, 2022 agreement in the PY2022 enrollment file had renewed early.[1] **Early renewers** (29 ACOs) were compared with **stayers** (277

ACOs), those in the same agreement in PY2022. The 28 ACOs absent from the PY2022 file (left after PY2021) were described but entered no registered test (Figure 1).

Variables

Expected gain from a reset. CMS does not publish what an early renewer's benchmark would have been had it not renewed, nor a stayer's had it reset. Following 42 CFR 425.601 as in force on January 1, 2022,[2] the registered plan rebuilt it in 6 steps for each of the 4 enrollment types (end-stage renal disease [ESRD], disabled, aged/dual eligible, aged/non-dual eligible): (1) the equally weighted mean of the ACO's per-capita spending in PY2019-PY2021, restated to PY2021 case mix and trended to 2021 dollars with the national-regional growth blend; (2) risk-standardized county spending averaged with weights equal to the ACO's assigned person-years by county in 2021 (the published proxy for the benchmark year's assignment) and adjusted to the ACO's case mix;[30] (3) the regional adjustment, $w \times (\text{regional} - \text{rebased})$, with the phase-in weight w and the cap of 425.601; (4) the approximate new benchmark, combining types by the ACO's PY2021 person-year shares; and (5) the expected gain, $100\times$ (new $-$ current) / current, where the current benchmark is the PY2021 updated per-capita benchmark. The gain splits into an **own-spending component** (rebased spending relative to the current benchmark, which is close to a restatement of the savings rate) and a **regional component** (the adjustment). Writing the adjustment out shows that, before capping,

$$\text{regional component} = w \times (\text{pull} - \text{own}),$$

so the regional component is also tied to past savings. Step (6) is the primary measure, the **regional pull**,

$$\text{regional pull} = 100 \times \frac{\sum_e s_e R_e - B}{B},$$

where R_e is the case-mix-adjusted regional spending for enrollment type e, s_e the ACO's share of person-years of that type, and B its current per-capita benchmark. All 4 are in percentage points of the current benchmark. They use final PY2021 data, published after the 2022 renewal decision was made during 2021, so they are ex post measures of what ACOs could anticipate from CMS's quarterly reports and their own earlier results.

Validation before use. Several steps can only be approximated from public files. The plan therefore required the rebuilt benchmark to be validated against the 2022 benchmarks CMS published for regular renewers (ACOs whose scheduled agreements ended in 2021) before any expected-gain

CC BY 4.0

variable was estimated, with a pass rule of a Spearman correlation of at least 0.80 and a median absolute percentage error of at most 5%, and a calibration step if the slope of actual on rebuilt fell outside 0.9 to 1.1. Early renewers were excluded from the validation sample and the 2022 results file was filtered as it was loaded, so validation could not reveal their actual reset. A second check compared the own-spending part with CMS's own benchmark-year spending for the same ACOs; its error SD set the measurement-error SD used in the decision model.

Other variables. From the results files: the savings rate (CMS-reported savings as a share of the benchmark) in PY2020 and PY2021, assigned beneficiaries, and log per-capita spending, PY2020-PY2024. From the enrollment files: cohort, PY2021 track and BASIC level, revenue category, assignment method and agreement number. An ACO's PY2021 BASIC level reflects both the glide path's schedule and its own earlier choices: ACOs choose their entry level, can advance faster, and could pause the advance for 2021.

Suppressed values. Hidden cells needed for the expected gain were filled by registered rules in order: exact recovery of a single hidden count from totals, the mean over years in which a value was shown, a regional ratio for a type hidden in every year, and, for a hidden risk score, no restatement and the region's mean risk score. ACOs still missing a variable were excluded from that analysis and counted; the plan required reporting as a limitation any exclusion above 10% of either group.

Hypotheses

The registered hypotheses were: **H2b (primary)**, early renewers had a larger regional pull than stayers; **H2a (secondary)**, a larger own-spending component (close to H1 restated, so not counted as separate evidence); **H1 (descriptive)**, a difference in PY2020 savings rate, with no motive attributed to its sign; **H3**, among BASIC-track ACOs, the probability of renewing early rises with the regional pull, the own-spending component and a higher BASIC level; and **H4**, early renewal changed per-capita spending in PY2022-PY2024 (2-sided). H1 to H3 describe who chose to reset and are associations. H4 concerns a bundled package (a new benchmark, for almost all a move to ENHANCED, and new benchmark years) whose parts cannot be separated.

Statistical analysis

Group comparisons (H1, H2a, H2b). The estimate was the difference in means, early renewers minus stayers, within each cohort, averaged with weights proportional to the early renewers in each cohort. Its posterior was a Bayesian bootstrap[31] (10,000 draws of Dirichlet(1, …, 1) weights within each cohort-by-group cell), reported as the posterior mean, 95% equal-tailed credible interval (CrI) and posterior probability of a positive value. Significance was by stratified permutation: labels

were shuffled within cohort 10,000 times, and the 2-sided P value was $(1+\#\{|\hat{\theta}_{\text{perm}}| \geq |\hat{\theta}_{\text{obs}}|\})/(1+10{,}000)$.[32] When the P value and the CrI disagree, both are reported and the disagreement noted.

Decision model (H3). A Bayesian logistic regression of early renewal among PY2021 BASIC-track ACOs on the 3 core predictors and 5 adjustment factors (cohort, log assigned beneficiaries, high revenue, prospective assignment, later agreement) was fit in PyMC[33] with the No-U-Turn sampler[34] (4 chains of 2,000 tuning and 2,000 retained draws, target acceptance 0.95). Continuous predictors were centered and divided by 2 SD[35] and binary predictors centered, so each coefficient is a log-odds difference per 2 SD or per category; coefficients had Normal(0, 1) priors. The own-spending component entered as a latent true value observed with normal error, with the error SD fixed by the validation step. Convergence required $\hat{R} \leq 1.01$, bulk and tail effective sample sizes of at least 400 and no divergences,[36] with one registered rerun; a Firth penalized regression[37,38] would have decided H3 had the model not converged. Because shuffling labels in a multi-predictor model tests the joint null, each core predictor was tested by direct simulation under the reduced model: the model without that predictor was fit, 10,000 outcome vectors were simulated from its posterior, the full model was refit to each, and the observed coefficient was compared with that null distribution. A fit with Student-t(3, 0, 2.5) priors, after Gelman and colleagues,[39] was the registered sensitivity analysis, with any differing core conclusion to be flagged.

Spending after renewal (H4). For each ACO, Δ was the mean log per-capita spending over its included years in PY2022-PY2024 minus the mean in PY2020-PY2021, and H4 applied the group-comparison estimator to Δ, reported as $100 \times (e^{\text{estimate}} - 1)$, the percent difference in spending growth. Log differences are also given in log points (1 log point = 0.01 in log spending, about 1%). The comparison group was ACOs not yet renewed: a stayer's years after it renewed into a 2023 or 2024 agreement were excluded, following the not-yet-treated design of Callaway and Sant'Anna.[40] With one pre-period change (PY2020 to PY2021, both affected by COVID-19) and about 29 treated ACOs, a nonsignificant pre-trend test is weak evidence of parallel trends,[41] so the plan replaced it with an equivalence condition in the spirit of equivalence testing:[42] H4 would be read as an effect only if the larger absolute bound of the pre-trend CrI, scaled by the mean distance between period midpoints, was smaller than the absolute H4 estimate. A trend-adjusted estimate (the pre-period slope carried forward, in the spirit of Rambachan and Roth[43]) and risk-adjusted, balanced-panel, propensity-weighted[44] and single-cohort estimates were reported beside it.

Power. Minimum detectable effects (MDEs) at 80% power and 2-sided $\alpha = .05$ for 29 vs 277 ACOs were computed after registration and committed before any confirmatory estimate: from the cross-ACO SD of the savings rate in PY2016-PY2018 (H1), the SD of each constructed variable among stayers, computed after validation (H2a, H2b), and the SD of the pre-period spending change (H4). For H3, detection was simulated at a log-odds difference of 0.5.

CC BY 4.0

Decision rules. Fixed in the plan before any result. For H2b, $P < .05$ with a positive difference was to be reported as "early renewers' regions spent more relative to their current benchmarks, evidence for the reset motive beyond past savings," $P < .05$ with a negative difference as evidence against it, and otherwise as no detectable difference. H2a used the same wording; H3 predictors were supported, contradicted or undetermined by their direct-simulation P values; and H4 was reported as no detectable difference, grew less or grew more, read as an effect only if the equivalence condition held. No test here could rule the sharing-rate motive in or out. As registered, there is one primary test, and the 3 H3 tests and the H4 test are not adjusted for multiplicity.

Covariate-adjusted estimates, conventional methods and robustness. Reported beside the registered methods and not used for decisions: covariate-adjusted estimates for H1, H2a and H2b from robust Bayesian linear regressions (Student-t errors on the standardized variable, Normal(0, 1) priors) on early renewal and cohort, or cohort, track, level and size; Welch t tests,[45] ordinary least squares with HC2 errors[46] and maximum-likelihood logistic regression in statsmodels;[47] 2-way fixed effects with an event study and errors clustered by ACO for H4; and the registered robustness checks: differences in medians, the January 1, 2020 cohort alone, PY2021 savings rates, the Firth model, benchmark years 2-3 only, no measurement-error model, complete cases and unadjusted county spending. Variables were built in dbt[48] on DuckDB;[49] the environment was locked with uv;[50] MCMC diagnostics used ArviZ;[51] every refit had its own seed from a fixed NumPy seed sequence.[52]

Deviations. All 7 analysis stages ran with code identical to the registered commit. After the last stage the lockfile took security updates to 2 packages (filelock and virtualenv); no result was produced with it, and it is logged as the one deviation. One sentence of the plan's background is imprecise: it describes ENHANCED as sharing 75% of savings and losses, whereas it shares 75% of savings and up to 75% of losses.

Ethics

The study used only publicly available, aggregate data on organizations and counties published by CMS. It involved no individual-level data, no interaction with people and no identifiable private information, so it was not human participants research, and ethics committee or institutional review board approval and informed consent were not required.

Results

Validation and sample

The rebuilt benchmark passed validation (eTable 1 in Supplement 1). For 101 of the 111 eligible regular renewers, it had a Spearman correlation of 0.94 with the benchmark CMS published and a median absolute percentage error of 1.6%; the slope of actual on rebuilt was 0.944, inside the registered range, so no calibration was applied. The own-spending part agreed with CMS's benchmark-year spending (Spearman 0.95; median error 1.4%), and its error SD set the measurement-error SD at 4.38 percentage points. That SD exceeds the own-spending component's spread among stayers (SD 3.3). Under a classical error model it could not, so the validation errors are probably heavy-tailed (a normal error with that SD would have a median absolute value near 3, against the 1.4% observed) or do not transfer exactly to the study sample. By the registered error model, the component is mostly noise at the level of one ACO; the savings rates, which CMS reports directly, are the firmer evidence of how far ACOs were beating their benchmarks.

Early renewers were more often in the January 1, 2020 cohort and at BASIC's two-sided levels C to E in PY2021 than stayers (Table 1); 16 of the 27 BASIC-track early renewers were at the highest level, E. Of those 27, 26 moved to ENHANCED in their new agreements, and no stayer moved to ENHANCED. The expected-gain variables could not be built for 3 early renewers (10.3%) and 36 stayers (13.0%), both above the plan's 10% line. A diagnostic query run after the results, outside the registered pipeline, found that in every one of these 39 ACOs CMS had suppressed the count of assigned ESRD beneficiaries in every county of the 2021 assignment file, so the ESRD part of regional spending had no county weights (for 15, the aged/dual counts were hidden too). The plan's fill rules cover hidden cells in the results files, not fully hidden county weights, so these ACOs were excluded and counted, as registered. The same query showed them to be smaller ACOs (median 6,702 and 8,578 assigned beneficiaries for excluded early renewers and stayers, against 10,757 and 15,649 for those included). One early renewer and 3 stayers had hidden values filled by the registered rules.

Table 1. Characteristics of the 334 ACOs in the 2019 and 2020 agreement cohorts, by group

Characteristic	Early renewers	Stayers	Left after PY2021
ACOs, No.	29	277	28
Agreement start, No. (%)			
July 1, 2019	11 (38)	159 (57)	19 (68)
January 1, 2020	18 (62)	118 (43)	9 (32)

Characteristic	Early renewers	Stayers	Left after PY2021
Track in PY2021, No. (%)			
BASIC levels A-B (one-sided)	7 (24)	133 (48)	23 (82)
BASIC levels C-E (two-sided)	20 (69)	76 (27)	4 (14)
ENHANCED	2 (7)	68 (25)	1 (4)
Assigned beneficiaries, PY2020, mean (SD), %			
Aged 85 y or older	10.9 (3.7)	12.4 (4.1)	13.3 (5.8)
Female	56.8 (1.9)	57.1 (2.3)	57.3 (2.5)
Savings rate, PY2020, pp			
Mean (SD)	6.4 (3.6)	4.6 (3.9)	2.7 (5.5)
Median	5.8	4.3	1.5
Regional pull, pp of benchmark			
ACOs with the variable, No.	26	241	20
Mean (SD)	5.0 (5.0)	1.8 (4.6)	4.8 (6.7)
Median	3.5	1.4	2.0
Own-spending component, pp of benchmark			
ACOs with the variable, No.	26	241	20
Mean (SD)	−5.3 (4.4)	−3.2 (3.3)	−1.8 (4.6)
Median	−4.2	−2.9	−1.0
Regional component, pp of benchmark			
ACOs with the variable, No.	26	241	20
Mean (SD)	3.7 (1.6)	2.2 (1.8)	2.0 (2.3)
Median	3.8	2.1	1.3

Abbreviations: ACO, accountable care organization; pp, percentage points; PY, performance year. Early renewers ended their agreements and began new ones on January 1, 2022; stayers remained in their agreements; leavers left the program after PY2021. The savings rate is CMS's reported savings as a share of the benchmark (positive when spending was below the benchmark). The expected-gain variables could not be built for every ACO; a diagnostic query run after the results, outside the registered pipeline, traced each exclusion to county counts of assigned beneficiaries with end-stage renal disease that were all suppressed (Results). Leavers were not compared in any registered test.

CC BY 4.0

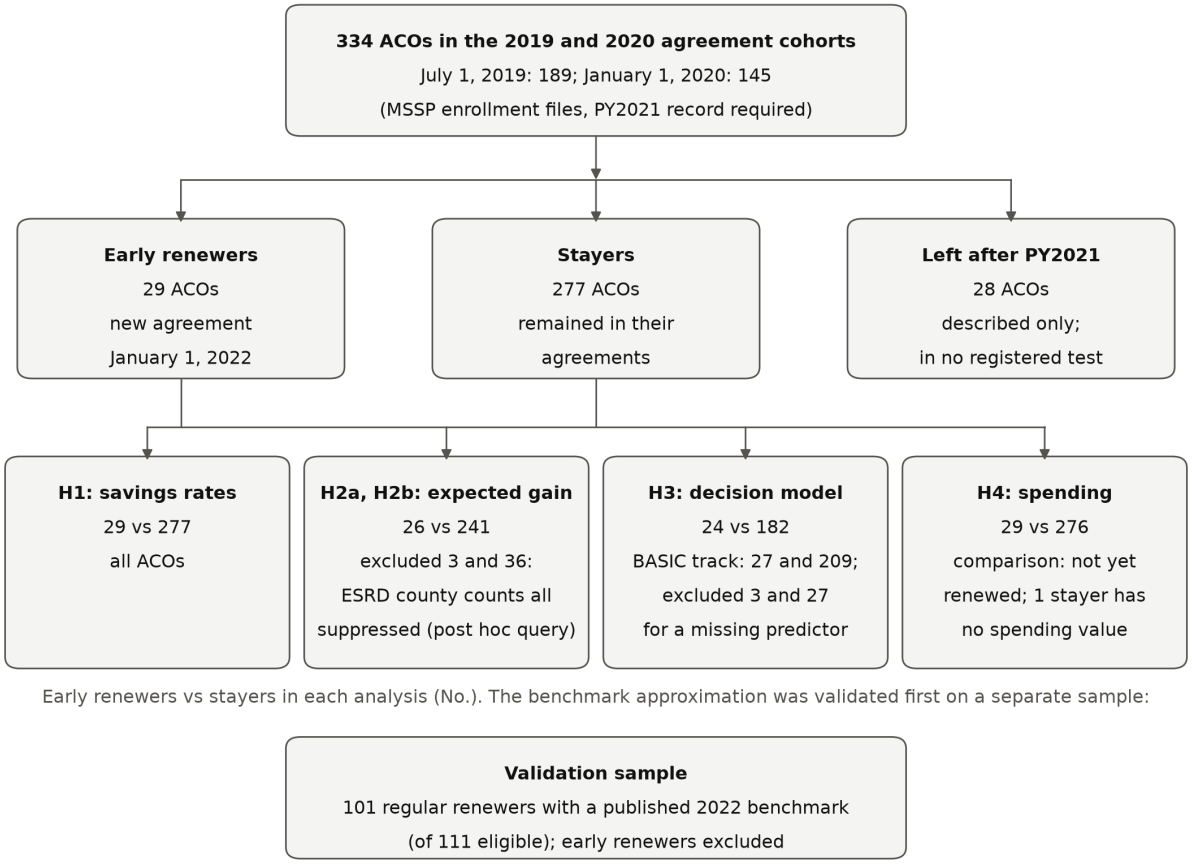


Figure 1. Flow of ACOs into each analysis. Counts are early renewers vs stayers. The reason for the expected-gain exclusions comes from a diagnostic query run after the results, outside the registered pipeline. ACO, accountable care organization; ESRD, end-stage renal disease; PY, performance year.

Who renewed early

Relative to their current benchmarks, early renewers' case-mix-adjusted regional spending was 3.4 percentage points higher than stayers' (95% CrI, 1.7 to 5.4; $P < .001$), above the MDE of 2.5 computed for 29 vs 277 ACOs, which is slightly understated for the 26 vs 241 compared (Table 2 and Figure 2). By the registered rule, H2b was supported: early renewers' regions spent more relative to their current benchmarks, evidence for the reset motive beyond past savings. At the median PY2021 benchmark of the study's ACOs, $11,312 per beneficiary per year, the difference of 3.35 points is about $379 per beneficiary-year of regional spending relative to the benchmark; the difference in the regional component itself, what a reset would add, was 1.6 points, about $176.

The own-spending component pointed the other way (−2.1 points; 95% CrI, −3.8 to −0.5; P = .003): early renewers' recent spending was lower relative to their benchmarks, contrary to the reset motive by the rule's wording. Early renewers also had higher savings rates in PY2020 (1.9 points; 95% CrI, 0.7 to 3.3) and PY2021 (1.6 points; 95% CrI, 0.2 to 3.1). The two parts of the gain offset: the regional component was larger for early renewers (1.6 points; 95% CrI, 0.9 to 2.2), the own-spending component smaller, and the total expected gain did not differ (−0.5 points; 95% CrI, −2.0 to 0.8; P = .33). In unweighted means the own-spending and regional parts summed to −5.3 + 3.7 = −1.6 points for early renewers and −3.2 + 2.2 = −1.0 for stayers: by the approximation, a reset would on average have lowered both groups' benchmarks.

Table 2. Early renewers compared with stayers: regional pull (H2b), own-spending component (H2a), savings rates (H1) and the expected gain

Measure (role)	Early / stayers, No.	Difference (95% CrI)	Pr(>0)	*P*	MDE	Verdict
H2b, regional pull (primary)	**26 / 241**	**+3.4 (1.7 to 5.4)**	**1.000**	**<.001**	**2.5**	**Supported**
H2a, own-spending component (secondary)	26 / 241	−2.1 (−3.8 to −0.5)	0.006	.003	1.8	Contradicted
H1, savings rate, PY2020 (descriptive)	29 / 277	+1.9 (0.7 to 3.3)	1.000	.01	2.7	
H1, savings rate, PY2021 (descriptive)	29 / 277	+1.6 (0.2 to 3.1)	0.988	.03	2.7	
Total expected gain (descriptive)	26 / 241	−0.5 (−2.0 to 0.8)	0.233	.33		
Regional component (descriptive)	26 / 241	+1.6 (0.9 to 2.2)	1.000	<.001		

Abbreviations: CrI, credible interval; MDE, minimum detectable effect at 80% power, two-sided α = .05, computed before the tests for 29 vs 277 ACOs (for the 26 vs 241 compared on the expected-gain variables it is 2.6 for H2b and 1.9 for H2a); Pr(>0), posterior probability that the difference is positive; pp, percentage points; PY, performance year. Differences are early renewers minus stayers within agreement cohort, averaged with weights proportional to early renewers per cohort. Posterior means and 95% equal-tailed CrIs are from a Bayesian bootstrap (10,000 draws); P values are from stratified permutation (10,000 shuffles within cohort). Expected-gain variables are in percentage points of the current benchmark; savings rates in percentage points. Verdicts follow the registered decision rules; H1 and the two descriptive rows carry none.

CC BY 4.0

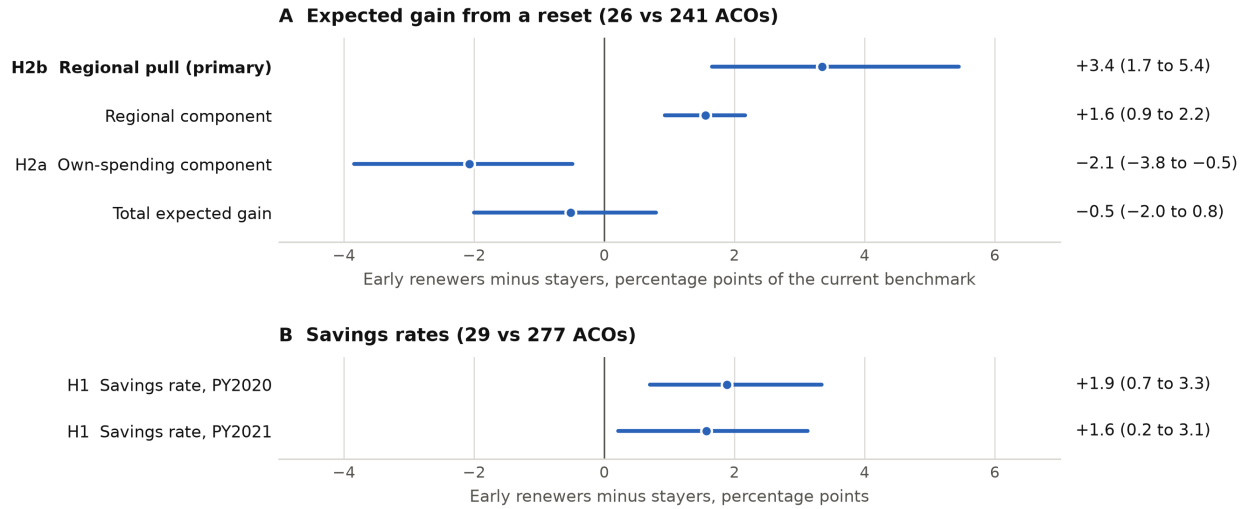


Figure 2. Early renewers minus stayers. A, Expected-gain variables, in percentage points of the current benchmark. B, Savings rates, in percentage points. Points are posterior means and lines 95% credible intervals from the Bayesian bootstrap.

Decision model

Among 206 BASIC-track ACOs (24 early renewers), early renewal was associated with a higher regional pull (log-odds difference per 2 SD, 1.34; 95% CrI, 0.44 to 2.30; direct-simulation P = .002) and a higher BASIC level (log-odds difference for levels C-E vs A-B, 1.34; 95% CrI, 0.38 to 2.31; P = .003); both predictions were supported (Table 3 and Figure 3). A log-odds difference of 1.34 multiplies the odds of renewing early by about 3.8. The own-spending component was undetermined (−0.05; 95% CrI, −2.01 to 1.92; P = .19); its interval is the prior's 95% range, because its measurement error exceeds its spread and the model could learn almost nothing about it. Without the measurement-error layer it was −0.64 (95% CrI, −1.51 to 0.23). The model converged on its first fit (largest $\hat{R}$ 1.001, no divergences), and the observed 24 early renewers fell inside the posterior predictive 95% range of 14 to 37 (eTable 5 in Supplement 1). Under the Student-t prior, which converged on its registered rerun, the pull (1.65; 95% CrI, 0.53 to 2.87) and BASIC-level (1.63; 95% CrI, 0.53 to 2.85) intervals also excluded zero, so no core conclusion differs between priors. The power analysis gave detection rates of only 17% (pull) and 14% (BASIC level) at a log-odds difference of 0.5, so estimates that pass the test are likely to overstate the true effects;[53] the directions are the result, and coefficients near 1.3 are probably too large.

Table 3. Decision model of early renewal among BASIC-track ACOs (H3)

Predictor	Posterior mean (95% CrI)	Pr(>0)	*P*	Verdict	Student-t prior (95% CrI)	Conventional OR (95% CI)
Regional pull	1.34 (0.44 to 2.30)	0.999	.002	Supported	1.65 (0.53 to 2.87)	4.01 (1.20 to 13.40)
Own-spending component	−0.05 (−2.01 to 1.92)	0.480	.19	Undetermined	−0.56 (−9.98 to 7.36)	0.53 (0.19 to 1.44)
Higher BASIC level (C-E)	1.34 (0.38 to 2.31)	0.998	.003	Supported	1.63 (0.53 to 2.85)	4.79 (1.47 to 15.59)
2020-01-01 cohort	0.63 (−0.24 to 1.51)	0.920		Adjustment	0.73 (−0.26 to 1.76)	1.97 (0.71 to 5.49)
Log assigned beneficiaries	0.45 (−0.49 to 1.36)	0.827		Adjustment	0.66 (−0.43 to 1.77)	2.00 (0.65 to 6.15)
High revenue	−0.53 (−1.69 to 0.59)	0.182		Adjustment	−0.55 (−2.10 to 0.85)	0.63 (0.14 to 2.89)
Prospective assignment	−0.77 (−1.79 to 0.19)	0.058		Adjustment	−1.01 (−2.27 to 0.15)	0.37 (0.11 to 1.31)
Later agreement	−0.67 (−1.62 to 0.31)	0.089		Adjustment	−0.91 (−2.04 to 0.22)	0.38 (0.12 to 1.19)

Abbreviations: CrI, credible interval; OR, odds ratio; Pr(>0), posterior probability that the coefficient is positive. Bayesian logistic regression of early renewal among 206 BASIC-track ACOs (24 early renewers; 3 early renewers and 27 stayers excluded for a missing predictor). Continuous predictors are centered and divided by 2 SD, so their coefficients are log-odds differences per 2 SD; binary predictors are centered, so theirs are per category (the BASIC level compares levels C-E with A-B). Core predictors (bold) are tested by direct simulation under the reduced model (10,000 simulated datasets); adjustment factors are estimates only. The own-spending component enters with a measurement-error model ($\sigma_v = 4.38$ pp). The Student-t column is the registered prior sensitivity analysis; the maximum-likelihood ORs are the conventional counterpart, reported only.

CC BY 4.0

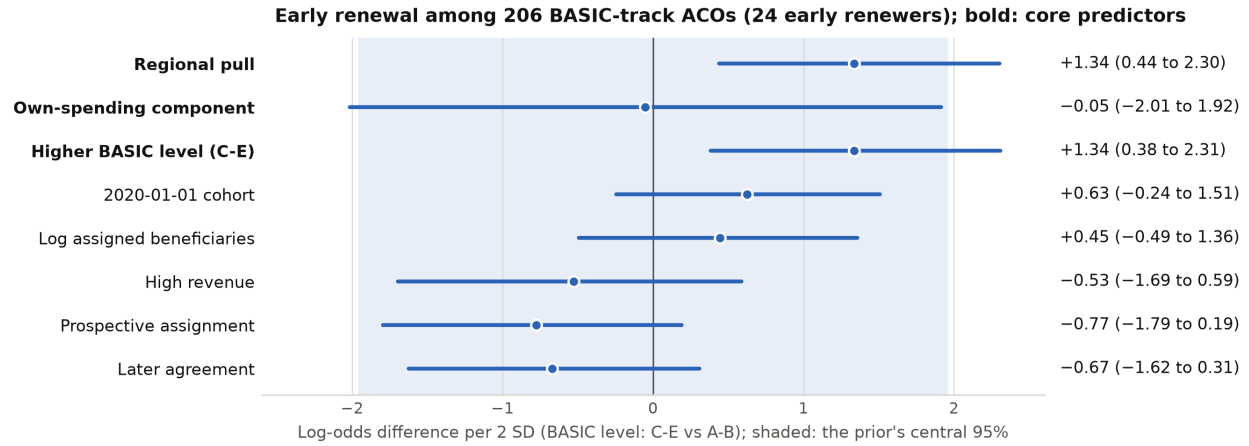


Figure 3. Decision model of early renewal among BASIC-track ACOs (H3). Points are posterior means and lines 95% credible intervals; the shaded band is the Normal(0, 1) prior's central 95%. Bold labels mark the core predictors.

Spending after renewal

Spending growth from PY2020-PY2021 to PY2022-PY2024 did not differ detectably between 29 early renewers and 276 comparison ACOs not yet renewed (−0.4%; 95% CrI, −4.0% to 2.5%; P = .78; MDE 3.8 log points) (Table 4 and Figure 4). The pre-trend estimate was 0.7% (95% CrI, −2.3% to 4.9%; P = .58), not evidence against parallel trends, but the equivalence condition did not hold: the scaled pre-trend bound, 2.41 years times 4.8 log points, was 11.5 log points (about 12%), far above the absolute H4 estimate of 0.4 log points. The condition could have held only for an estimate above about 12% in absolute value, 3 times the MDE, so with one pre-period it could not support reading an estimate of plausible size as an effect, and a trend difference of the size the pre-period allows could hide an effect in either direction. H4 is therefore reported as a difference in spending changes that cannot be told apart from a continuation of a pre-period difference in trends. Under parallel trends, the interval rules out a drop in spending growth of more than about 4% or a rise of more than about 2.5%; the trend-adjusted interval (−12.0% to 2.7%) is too wide to inform.

Attrition was uneven: by PY2024, 115 of 277 stayers (42%) had left the spending panel (72 left the program, 42 renewed and 1 had no spending value), against 4 of 29 early renewers (14%) (eTable 8 in Supplement 1). The balanced panel of ACOs with all 5 years gave −1.6% (95% CrI, −5.6% to 1.6%; P = .30).

Table 4. Spending growth after early renewal (H4), with the pre-trend and robustness estimates

Estimate	Early / comparison, No.	Difference, %	95% CrI	*P*
H4, difference-in-differences	**29 / 276**	**−0.4**	**−4.0 to 2.5**	**.78**
Pre-trend, PY2020 to PY2021	29 / 276	+0.7	−2.3 to 4.9	.58
Trend-adjusted	29 / 276	−4.0	−12.0 to 2.7	.14
Risk-adjusted outcome	29 / 276	+0.3	−2.0 to 2.5	.85
Balanced panel (all 5 years)	25 / 162	−1.6	−5.6 to 1.6	.30
Propensity-weighted	305 total	−0.5	NA	.67
2020-01-01 cohort only	18 / 118	−2.9	−8.3 to 1.3	.13
Two-way fixed effects (conventional; 95% CI)	306 ACOs	−1.5	−5.4 to 2.4	.44

Abbreviations: CrI, credible interval; NA, not available (the propensity-weighted estimate has a permutation P value but no interval; its largest weight is 0.83); PY, performance year. The outcome is the change in log per-capita spending from PY2020-PY2021 to PY2022-PY2024, shown as a percent difference in spending growth, early renewers minus ACOs not yet renewed. A stayer's years after it renewed into a 2023 or 2024 agreement are excluded. Equivalence condition: the pre-trend bound scaled to the distance between period midpoints (2.41 years) is 11.5 log points, above the absolute H4 estimate of 0.4 log points (1 log point = 0.01 in log spending, about 1%), so the condition does not hold and H4 is not read as an effect. The two-way fixed-effects model has one more ACO: the stayer with no post-period spending value keeps its pre-period years.

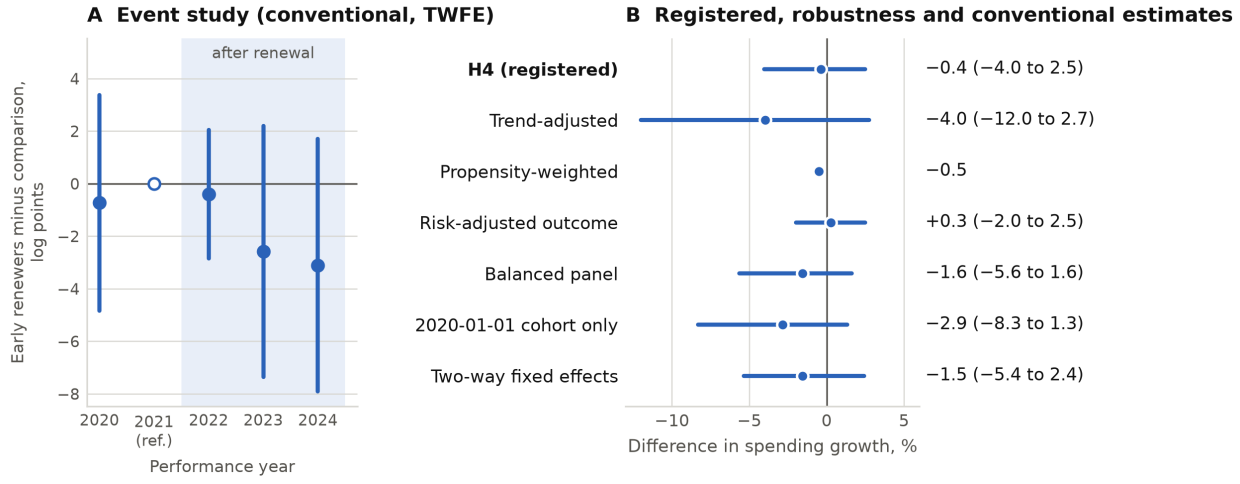


Figure 4. Spending after early renewal. A, Event study from the conventional 2-way fixed-effects model, in log points relative to PY2021, with 95% CIs clustered by ACO. B, The registered H4 estimate and the robustness and conventional estimates, as the percent difference in spending growth with 95% intervals (the propensity-weighted estimate has no interval); sample sizes are in Table 4. TWFE, 2-way fixed effects.

CC BY 4.0

Robustness and conventional methods

Every registered verdict agreed with its conventional counterpart (eTable 7 in Supplement 1). For H2b, ordinary least squares with HC2 errors gave 3.3 points (95% CI, 1.3 to 5.4; $P = .001$), and for H3 the maximum-likelihood odds ratios were 4.0 (95% CI, 1.2 to 13.4) for the regional pull and 4.8 (95% CI, 1.5 to 15.6) for the BASIC level; 2-way fixed effects gave −1.5% (95% CI, −5.4% to 2.4%) for H4. H2b kept its direction in every robustness check (eTable 3 in Supplement 1), but its size depended on the estimator: the difference in medians was 2.8 points, and the robust (Student-t) regressions gave 2.1 (95% CrI, 0.5 to 3.8) with cohort and 2.2 (95% CrI, 0.7 to 3.8) with track, level and size added, below the MDE of 2.5, so a few large values among early renewers carry part of the difference in means; without the case-mix adjustment of regional spending the difference was 6.9. H2a and H1 kept their direction but were weaker by medians (H2a, $P = .09$; H1, $P = .06$) and in the adjusted regressions, whose intervals reached zero (H2a, −2.6 to 0.02; H1, −0.1 to 2.6). In H3 the BASIC-level interval stayed above zero in every check, while the pull interval reached zero with the 2020 cohort alone (1.15; 95% CrI, −0.02 to 2.39) and with unadjusted county spending (0.75; 95% CrI, −0.02 to 1.56) (eTable 4 in Supplement 1).

Among the exploratory analyses (eTables 9-11 in Supplement 1), the 28 ACOs that left after PY2021 had the lowest PY2020 savings rates (mean 2.7 points, against 6.4 for early renewers and 4.6 for stayers) and, for the 20 with the variable, a mean regional pull similar to early renewers' (4.8 vs 5.0; medians 2.0 vs 3.5) (Table 1). The 2 early renewers already in ENHANCED, whose renewal could not raise the sharing rate, fell between the 26th and 68th percentiles of stayers on every variable.

Discussion

In this preregistered study of the 2019 and 2020 MSSP cohorts, ACOs that ended their agreements early to begin new ones on average had regions that spent more relative to their current benchmarks, and, among BASIC-track ACOs, were more often already at the higher, two-sided levels. They were also beating their benchmarks by more than ACOs that stayed, so by the study's approximation a reset would not on average have raised their benchmarks: the regional gain and the own-spending loss roughly offset. Spending growth in the 3 years after renewal did not differ detectably from that of ACOs not yet renewed, a result the design could not separate from a continuation of a pre-period difference in trends.

Comparison with earlier work

The primary result extends to a new decision the pattern that participation follows the benchmark formula. Participation shifted toward providers whose spending was low relative to their region once

regional spending entered benchmarks,[11] and ACOs whose effective benchmarks fell were more likely to leave.[12] Here the choice is when to reset rather than whether to join or leave, and early renewers had, on average, more room under their regions' spending. The finding that early renewers were savers fits reports that ACOs earning bonuses survived longer in the program[14] and, with borderline precision, that ACOs that had achieved savings were more likely to stay after the 2019 rules,[16] but the same success meant that the own-spending part of a reset would have cut their benchmarks, so the simple account that ACOs reset to get a higher benchmark does not describe them on average. That the size of the regional difference doubled without case-mix adjustment is consistent with evidence that comparisons of ACOs with their regions depend heavily on risk adjustment.[10] The concern that rebasing rewards higher spending before a new agreement[9] could be neither confirmed nor excluded here, and the costs to Medicare of early renewal, which earlier work estimated for departures,[18] were not estimated.

The association with a higher BASIC level bears on the sharing-rate motive at least as much as on the reset motive. In these data renewing early and moving to ENHANCED were one choice: 26 of the 27 BASIC-track early renewers moved to ENHANCED, and no stayer did. ACOs already bearing two-sided risk at Level E faced a short step to ENHANCED's 75% sharing rate. Because ACOs partly choose their BASIC level, the association may also reflect an appetite for risk rather than only proximity to ENHANCED. A high regional pull is also a plausible reason to want the higher rate, because an ACO with room under its region's spending may expect larger savings to share. The 2 early renewers already in ENHANCED, for whom only the reset could change, looked like typical stayers on every variable, but 2 ACOs cannot separate the motives. No test here can rule either motive in or out.

Implications

For program design, the results suggest that the early-renewal option let ACOs choose when to reset, and that those that used it were, by the approximation, positioned to gain from the regional adjustment rather than from a higher benchmark overall, with a higher sharing rate available to almost all of them. That is selection into an option the program created; it is not evidence that renewing changed spending. The rules have since changed. For agreement periods beginning January 1, 2024 or later, an ACO that saved in its benchmark years can receive a prior savings adjustment, and its benchmark gets the higher of that adjustment and the regional adjustment; negative regional adjustments are partly offset; and the annual update blends in a fixed projected trend.[54] Under these rules a reset costs an ACO that has been beating its benchmark less, so the incentive to renew early, and which ACOs do so, may differ. Two questions remain open: what early renewal cost Medicare, and how much of the choice reflected the sharing rate rather than the benchmark, which a period in which ACOs could move to ENHANCED without resetting would

CC BY 4.0

separate.

Limitations

This study has limitations. First, the motives are bundled, and the regional pull, though it has no term for benchmark-year spending, is not free of the ACO's own data: its benchmark was set from earlier spending, and regional spending is adjusted with the ACO's own PY2021 risk scores, so a high pull partly measures how low-cost the ACO had been relative to its region and may carry differences in coding intensity. The case-mix adjustment has no published counterpart to validate it. Second, the expected gain is an approximation built from final PY2021 data that ACOs did not have when they decided; validation bounds its error in benchmark levels, but the own-spending component's estimated error exceeds its spread, so H2a and the total gain are weak evidence for any single ACO, and the regional part could not be checked. Third, with 29 early renewers, 24 in the decision model, H3 was underpowered and its supported coefficients are probably overestimates. Fourth, exclusions exceeded the plan's 10% line in both groups, and the excluded ACOs were smaller, so H2a, H2b and H3 describe the larger ACOs. Fifth, H4 had one COVID-19-affected pre-period; uneven attrition left the comparison group's later years increasingly selected; the decision was made during 2021, the last pre-period year and a benchmark year of the new agreement, so anticipation is possible; and spending can move through changes in participant lists or assignment method (which changed at PY2022 for 2 of 29 early renewers and 14 of 277 stayers) rather than through behavior. Sixth, the base population required a PY2021 enrollment record, and leavers, a third choice with a mean pull similar to early renewers', were set aside by comparing early renewers only with stayers. Seventh, the results describe the rules for agreements beginning in 2022. Finally, the analysis reads its table without a fixed row order, so a rerun reproduces deterministic values exactly but resampled and simulated values only within simulation error (for example, H1's simulated power at the MDE was 80.4% as committed and 77.6% in a rerun from a fresh clone), far from changing any verdict.

Conclusions

ACOs that renewed early into 2022 MSSP agreements more often had regions that spent more relative to their benchmarks and were more often at higher BASIC levels, for which renewal opened the ENHANCED track. By the study's approximation, a reset would not on average have raised their benchmarks, and spending growth afterward showed no detectable difference. These are associations under the 2022 rules, measured with public data under a preregistered plan.

CC BY 4.0

Declarations

Data availability: All data are public CMS files.[21,25–27] The list of the 40 files downloaded, with sizes, SHA-256 hashes and URLs, and a script that downloads and verifies them are in the study repository;[24] an archived copy of the snapshot and the results notebook exports are in the OSF project.[28] The committed results files, from which every table and figure here is generated, are in the study repository and in this manuscript's repository (https://github.com/stharrold/mssp-preprint-2026).

Code availability: The registered analysis code (commit a4064f5), its tests and step-by-step instructions to rerun and adapt the analysis are at https://github.com/stharrold/stharrold.github.io (directory analysis/mssp-benchmark-ratchet).[24] A longer account of the study for a general audience was published online.[55]

Funding: This study received no funding.

Competing interests: The author declares no competing interests.

Author contributions: STH conceived and designed the study, registered the plan, acquired the data, ran the analysis, interpreted the results, drafted and revised the manuscript, and is accountable for all aspects of the work.

Use of artificial intelligence: The author chose the question, hypotheses and decision rules, approved and registered the plan, ran the analysis, and is responsible for the results and this manuscript. Claude Code (Claude Opus 5.5; Anthropic) drafted the analysis code, the scripts that generate this manuscript's tables and figures, and parts of the text under the author's direction, and separate instances reviewed the plan, the code and the manuscript. The author reviewed and edited all output and verified the numbers against the results files.

References

1. Centers for Medicare & Medicaid Services. 42 CFR part 425: Medicare Shared Savings Program. Published online 2022. https://www.ecfr.gov/on/2022-01-01/title-42/chapter-IV/subchapter-B/part-425

2. Centers for Medicare & Medicaid Services. 42 CFR 425.601: Establishing, adjusting, and updating the benchmark for agreement periods beginning on July 1, 2019, and in subsequent years. Published online 2022. https://www.ecfr.gov/on/2022-01-01/title-42/chapter-IV/subchapter-B/part-425/subpart-G/section-425.601

3. McWilliams JM, Hatfield LA, Chernew ME, Landon BE, Schwartz AL. Early Performance of Accountable Care Organizations in Medicare. *New England Journal of Medicine*. 2016;374(24):2357-2366. doi:10.1056/nejmsa1600142

4. McWilliams JM, Hatfield LA, Landon BE, Hamed P, Chernew ME. Medicare Spending after 3 Years of the Medicare Shared Savings Program. *New England Journal of Medicine*. 2018;379(12):1139-1149. doi:10.1056/nejmsa1803388

5. Bond AM, Civelek Y, Schpero WL, et al. Long-Term Spending of Accountable Care Organizations in the Medicare Shared Savings Program. *JAMA*. 2025;333(21):1897-1905. doi:10.1001/jama.2025.3870

6. McWilliams JM, Hatfield LA, Landon BE, Chernew ME. Savings or Selection? Initial Spending Reductions in the Medicare Shared Savings Program and Considerations for Reform. *The Milbank Quarterly*. 2020;98(3):847-907. doi:10.1111/1468-0009.12468

7. Markovitz AA, Hollingsworth JM, Ayanian JZ, Norton EC, Yan PL, Ryan AM. Performance in the Medicare Shared Savings Program After Accounting for Nonrandom Exit: An Instrumental Variable Analysis. *Annals of Internal Medicine*. 2019;171(1):27-36. doi:10.7326/m18-2539

8. Markovitz AA, Hollingsworth JM, Ayanian JZ, et al. Risk Adjustment In Medicare ACO Program Deters Coding Increases But May Lead ACOs To Drop High-Risk Beneficiaries. *Health Affairs*. 2019;38(2):253-261. doi:10.1377/hlthaff.2018.05407

9. Douven R, McGuire TG, McWilliams JM. Avoiding Unintended Incentives In ACO Payment Models. *Health Affairs*. 2015;34(1):143-149. doi:10.1377/hlthaff.2014.0444

10. Rose S, Zaslavsky AM, McWilliams JM. Variation In Accountable Care Organization Spending And Sensitivity To Risk Adjustment: Implications For Benchmarking. *Health Affairs*. 2016;35(3):440-448. doi:10.1377/hlthaff.2015.1026

11. Lyu PF, Chernew ME, McWilliams JM. Benchmarking Changes And Selective Participation In The Medicare Shared Savings Program. *Health Affairs*. 2023;42(5):622-631. doi:10.1377/hlthaff.2022.01061

CC BY 4.0

12. Chen AJ, McWilliams JM. How Benchmark Changes Affect Participation in Accountable Care Organizations: Prospects for Voluntary Payment Models. *American Journal of Health Economics*. 2025;11(1):38-62. doi:10.1086/726748

13. Srivastava A, Shay A, Kaufman SR, et al. Accountable Care Organization Efficiency on Entry and Shared Savings Bonuses. *JAMA Network Open*. 2026;9(2):e260166. doi:10.1001/jamanetworkopen.2026.0166

14. Bleser WK, Saunders RS, Muhlestein DB, McClellan M. Why Do Accountable Care Organizations Leave The Medicare Shared Savings Program? *Health Affairs*. 2019;38(5):794-803. doi:10.1377/hlthaff.2018.05097

15. Zhu X, Huang H, MacKinney AC, Ullrich F, Mueller K. Medicare accountable care organization characteristics associated with participation in 2-sided risk. *The Journal of Rural Health*. 2023;39(1):302-308. doi:10.1111/jrh.12672

16. Ying M, Hirth RA, Yan P, et al. Changes in Shared Savings Program Participation After Launch of Pathways to Success. *Journal of General Internal Medicine*. 2023;38(7):1780-1782. doi:10.1007/s11606-022-07947-x

17. Ying M, Forman JH, Murali S, et al. Factors affecting Accountable Care Organizations' decisions to remain in or exit the Medicare Shared Savings Program following Pathways to Success. *Health Affairs Scholar*. 2024;2(1):qxad093. doi:10.1093/haschl/qxad093

18. Moloci NM, Si Y, Norton EC, Ryan AM, Hollingsworth JM. Predicting Losses from Medicare Shared Savings Program Departures. *Journal of General Internal Medicine*. 2021;36(8):2490-2491. doi:10.1007/s11606-020-06424-7

19. Trombley MJ, McWilliams JM, Fout B, Morefield B. ACO Investment Model Produced Savings, But The Majority Of Participants Exited When Faced With Downside Risk. *Health Affairs*. 2022;41(1):138-146. doi:10.1377/hlthaff.2020.01819

CC BY 4.0

20. Centers for Medicare & Medicaid Services. Medicare program; Medicare Shared Savings Program; accountable care organizations: Pathways to Success and extreme and uncontrollable circumstances policies for performance year 2017. Final rule. Published online 2018. https://www.federalregister.gov/documents/2018/12/31/2018-27981/medicare-program-medicare-shared-savings-program-accountable-care-organizations-pathways-to-success

21. Centers for Medicare & Medicaid Services. Accountable care organizations: Shared Savings Program enrollment, performance years 2014-2026. Published online 2026. Accessed October 22, 2026. https://data.cms.gov/medicare-shared-savings-program/accountable-care-organizations

22. Harrold ST. Why did Medicare ACOs reset their benchmarks early? A pre-registered analysis of early renewal in the Medicare Shared Savings Program. Published online 2026. doi:10.17605/OSF.IO/C53W9

23. von Elm E, Altman DG, Egger M, et al. The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) Statement: Guidelines for Reporting Observational Studies. *PLoS Medicine*. 2007;4(10):e296. doi:10.1371/journal.pmed.0040296

24. Harrold ST. Registered analysis code for the MSSP early-renewal study (analysis/mssp-benchmark-ratchet), commit a4064f5. Published online 2026. https://github.com/stharrold/stharrold.github.io/tree/a4064f592a492d971ce3c27e95afb3050e6bc01d/analysis/mssp-benchmark-ratchet

25. Centers for Medicare & Medicaid Services. Performance year financial and quality results, performance years 2013-2024. Published online 2026. Accessed October 22, 2026. https://data.cms.gov/medicare-shared-savings-program/performance-year-financial-and-quality-results

26. Centers for Medicare & Medicaid Services. County-level aggregate expenditure and risk score data on assignable beneficiaries. Published online 2026. Accessed October 22, 2026. https://data.cms.gov/medicare-shared-savings-program/county-level-aggregate-expenditure-and-risk-score-data-on-assignable-beneficiaries

CC BY 4.0

27. Centers for Medicare & Medicaid Services. Number of accountable care organization assigned beneficiaries by county. Published online 2026. Accessed October 22, 2026. https://data.cms.gov/medicare-shared-savings-program/number-of-accountable-care-organization-assigned-beneficiaries-by-county

28. Harrold ST. Why did Medicare ACOs reset their benchmarks early? Project files: Results notebook exports and archived CMS data snapshot. Published online 2026. doi:10.17605/OSF.IO/Z4AN5

29. Centers for Medicare & Medicaid Services. CMS cell suppression policy. Published online 2020. https://web.archive.org/web/20250516214605/https://www.hhs.gov/guidance/document/cms-cell-suppression-policy

30. Centers for Medicare & Medicaid Services. Medicare Shared Savings Program: Shared savings and losses and assignment methodology specifications, version 10. Published online 2022. https://www.cms.gov/files/document/medicare-shared-savings-program-shared-savings-and-losses-and-assignment-methodology-specifications.pdf-1

31. Rubin DB. The Bayesian Bootstrap. *The Annals of Statistics*. 1981;9(1):130-134. doi:10.1214/aos/1176345338

32. Phipson B, Smyth GK. Permutation P-values Should Never Be Zero: Calculating Exact P-values When Permutations Are Randomly Drawn. *Statistical Applications in Genetics and Molecular Biology*. 2010;9(1):39. doi:10.2202/1544-6115.1585

33. Abril-Pla O, Andreani V, Carroll C, et al. PyMC: a modern, and comprehensive probabilistic programming framework in Python. *PeerJ Computer Science*. 2023;9:e1516. doi:10.7717/peerj-cs.1516

34. Hoffman MD, Gelman A. The No-U-Turn sampler: Adaptively setting path lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research*. 2014;15:1593-1623. https://jmlr.org/papers/v15/hoffman14a.html

35. Gelman A. Scaling regression inputs by dividing by two standard deviations. *Statistics in Medicine*. 2008;27(15):2865-2873. doi:10.1002/sim.3107

CC BY 4.0

36. Vehtari A, Gelman A, Simpson D, Carpenter B, Bürkner PC. Rank-normalization, folding, and localization: an improved R-hat for assessing convergence of MCMC. *Bayesian Analysis*. 2021;16(2):667-718. doi:10.1214/20-ba1221

37. Firth D. Bias reduction of maximum likelihood estimates. *Biometrika*. 1993;80(1):27-38. doi:10.1093/biomet/80.1.27

38. Luo J. firthmodels 0.8.2: Firth-penalized models in Python. Published online 2026. https://github.com/jzluo/firthmodels

39. Gelman A, Jakulin A, Pittau MG, Su YS. A weakly informative default prior distribution for logistic and other regression models. *The Annals of Applied Statistics*. 2008;2(4):1360-1383. doi:10.1214/08-aoas191

40. Callaway B, Sant'Anna PHC. Difference-in-Differences with multiple time periods. *Journal of Econometrics*. 2021;225(2):200-230. doi:10.1016/j.jeconom.2020.12.001

41. Roth J. Pretest with Caution: Event-Study Estimates after Testing for Parallel Trends. *American Economic Review: Insights*. 2022;4(3):305-322. doi:10.1257/aeri.20210236

42. Lakens D. Equivalence tests: a practical primer for t tests, correlations, and meta-analyses. *Social Psychological and Personality Science*. 2017;8(4):355-362. doi:10.1177/1948550617697177

43. Rambachan A, Roth J. A More Credible Approach to Parallel Trends. *Review of Economic Studies*. 2023;90(5):2555-2591. doi:10.1093/restud/rdad018

44. Rosenbaum PR, Rubin DB. The central role of the propensity score in observational studies for causal effects. *Biometrika*. 1983;70(1):41-55. doi:10.1093/biomet/70.1.41

45. Welch BL. The generalization of "Student's" problem when several different population variances are involved. *Biometrika*. 1947;34(1-2):28-35. doi:10.1093/biomet/34.1-2.28

46. MacKinnon JG, White H. Some heteroskedasticity-consistent covariance matrix estimators with improved finite sample properties. *Journal of Econometrics*. 1985;29(3):305-325. doi:10.1016/0304-4076(85)90158-7

CC BY 4.0

47. Seabold S, Perktold J. Statsmodels: Econometric and Statistical Modeling with Python. *Proceedings of the Python in Science Conference*. Published online 2010:92-96. doi:10.25080/majora-92bf1922-011

48. dbt Labs. dbt-core 1.12.6 and dbt-duckdb 1.11.0. Published online 2026. https://github.com/dbt-labs/dbt-core

49. Raasveldt M, Mühleisen H. DuckDB: an Embeddable Analytical Database. *Proceedings of the 2019 International Conference on Management of Data*. Published online June 25, 2019:1981-1984. doi:10.1145/3299869.3320212

50. Astral Software. uv: An extremely fast Python package and project manager. Published online 2026. https://github.com/astral-sh/uv

51. Kumar R, Carroll C, Hartikainen A, Martin O. ArviZ a unified library for exploratory analysis of Bayesian models in Python. *Journal of Open Source Software*. 2019;4(33):1143. doi:10.21105/joss.01143

52. Harris CR, Millman KJ, van der Walt SJ, et al. Array programming with NumPy. *Nature*. 2020;585(7825):357-362. doi:10.1038/s41586-020-2649-2

53. Gelman A, Carlin J. Beyond Power Calculations: Assessing Type S (Sign) and Type M (Magnitude) Errors. *Perspectives on Psychological Science*. 2014;9(6):641-651. doi:10.1177/1745691614551642

54. Centers for Medicare & Medicaid Services. 42 CFR 425.652, 425.656 and 425.658: Benchmarks, regional adjustment and prior savings adjustment for agreement periods beginning on January 1, 2024, and in subsequent years. Published online 2024. https://www.ecfr.gov/on/2024-01-01/title-42/chapter-IV/subchapter-B/part-425/subpart-G/section-425.652

55. Harrold ST. Why did Medicare ACOs reset their benchmarks early? A pre-registered study. Published online 2026. https://stharrold.github.io/20261027-preregistered-mssp-early-renewal.html

CC BY 4.0